

A Packing of 6 Four-Spaces in $\mathbb{R}^9$

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Abstract

We give 6 4-dimensional subspaces of $\mathbb{R}^9$ whose smallest squared chordal distance is at least 2.666497. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.592416, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 6 subspaces of dimension 4 in $\mathbb{R}^9$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.666497 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 9$, $n = 4$. The cell $N = 6$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.592416, obtained by deleting 1 of the 7 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(9 - 4)/9 \cdot 6/5 = 2.666667$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = Y G^{-1} Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m9n4N6.txt`

in <https://github.com/Ge-limin/math>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 61 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
133324873643798279246241410771413720735256926165297399929940278288891782
279408106157514975107183675622751278997563573013293200309513701580114524
1660123150042227545866521030301257032804153319321
```

denominator:

```
499999999997811519237540868577528945269179182902341856654165244631419427
871385309984370980530849477097386176633933858056779901581575927712827341
862797348772800155800941878717811774219616593668
```

B Coordinates

6 blocks of 9 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m9n4N6.txt`.

```
47672507812 -8784526258 577783834870 13520313481 -278811851125 -395809784107 -284125292519 290479161312321468836 -92635593671 219363036498 227027800392
256645274463 626918244294 125219733941 -588269585833335671164 -329903796413 -448663204654 -184697893732884753951 -154975574715 -221001759751 -290310181157
-262185788337 -22047478924 -186269055205 -186984079633333506842 205199368351 118060894165 -5936410774220742336900 210838034997 -11367850806 -760277437405
177996835463 192926468406 -165502465422 -11874482418234592090382 -667659155830 289988474793 -2815667457395995857992 3952271405 -325032179569 177554307466
393056762851 -43741645517 -263879391187 33566448480 426865690014 121937007859 -267570667119 672039437381993029774 346762914507 667342929771 170596663449
-356050977268 14317645324 -137640647953 290581323138065052893 -379817470186 372075555934 21160261764 659156212376 -31179550299 -397975031851 450975397003
9044362154 662120130703 171218706451 661395259206 -75839528294 185598402520 -271979194304 -579513997108951498910 772745053246 -75656706978 128576514220
-246097452021 326490234968 -634649068166 -32875321086856428631 -110217375516 -252697900512 4014263538826153797 145284246239 -112290391147 62269458286
699660346482 -148567715195 -260242320122 2839084871022119448058 202232513101 521612178167 1079891595916987242068 428338426739 -418567979510 -41708834953
-45002788224 -434491644788 -370764428713 3936210127753198622446 -174882207963 531759154395 597074344407255956158 634752185838 -261989637161 3704444362
733593550064 -2234513676 214049019516 -37293530485021793250725 291088104781 -152885966543 322658281457215692333 -29996819237 -644904683642 189992921741
-281176697408 -239076586785 330451753315 -361562006229086369006 68494031662 -370382203202 398322346735 104690474385 684399535311 330000914048 161811371578
-550341952614 32198038007 392944883915 -21483171721890493721626 226122541104 677588611951 -14622200862385495152103 323988510683 -98993486591 -377773098717
74064128935 565244316203 267752979894 529122510637 372487837594 465170543780 -41774578466 -51911605876 64035053928 136941438497 -239638948595 75134245568
-16115964911 501613475640 -39270975121 -23246575954720267249815 17448041955 149871177642 56112059588 374450396305 -33725132831 56116800939 597223137168
-179383061410 102997348671 -54405205121 1154063613916715272517 -494925053345 -227172068725 6587269264815932736335 27741304568 -317485223642 -168592050368
196616587862 -408046606673 598383478843 3047033087325474740541 -351880740778 49702881403 35543415805480659247528 -38145222651 332472652948 -511141426536
36014823624 -69209728818 -344755175095 -298208501531194637610391 493358043916 -138797582596 570879624202879158977 -26525351422 -353338664960 -378132337813
```